

# Empirical Proof Record: VALIDATED

2026-05-20T15:34:36.709536+00:00

## Empirical Proof Record — VALIDATED

**Proof ID:** 94966e2e44a5f0c30f07a752d76c7ca1...

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### 1. Claim

The posterior 94% HDI width on  $\mu_\Phi$  contracts at the theoretical  $\sqrt{N}$  rate as the number of observations grows. Specifically,  $\text{HDI\_width}(N=200) / \text{HDI\_width}(N=20) \leq 0.40$  (theoretical contraction factor  $\sqrt{(20/200)} = 0.316$ ; the 0.40 ceiling allows finite-sample slack).

- **Operationalisation:** Fit a PyMC NUTS posterior to subsamples of the  $\Phi$  trajectory at sizes 20, 50, 100, 200. Posterior model:  $\mu \sim \text{Normal}(\text{prior})$ ,  $\sigma \sim \text{HalfNormal}$ ,  $y_i \sim \text{Normal}(\mu, \sigma)$ . Compute the 94% HDI width on  $\mu$  at each size; the contraction metric is  $\text{HDI\_width}(N=200) / \text{HDI\_width}(N=20)$ .
- **Threshold:** `hdi_contraction_ratio <= 0.4 ratio`
- **H0:** `hdi_contraction_ratio > 0.40` (posterior fails to tighten at the predicted rate — data may be non-Gaussian, non-stationary, or sampler miscalibrated)
- **H1:** `hdi_contraction_ratio <= 0.40` (posterior tightens approximately at the theoretical  $\sqrt{N}$  rate, confirming Bayesian inference behaves correctly on the  $\Phi$  trajectory)

### 2. Verdict

**VALIDATED** — observed 0.27735050659795846

HDI width  $N=20 \rightarrow N=200$ : 0.0654  $\rightarrow$  0.0181; contraction ratio 0.2774 (theoretical 0.3162; ceiling 0.4000). Empirical  $\Phi$  mean=0.6140 sd=0.0680 (n=200).

### 3. Evidence

| Pillar              | Statistic             | Value               | 95% CI | Library    |
|---------------------|-----------------------|---------------------|--------|------------|
| pymc_nuts_posterior | hdi_contraction_ratio | 0.27735050659795846 | —      | pymc 6.0.0 |

### 4. Pre-registration

- Registered at: 2026-05-20T15:34:36.444141+00:00
- Config hash: a47e4fd070e7935308f57886af093806...
- Data hash: 9ef317fe291445319fcba522d29836dc...

### 5. Signature

28d5f4d5efce54d410ddb5febd225da389e7c4511c4898597cbf231ecc7e189b